

Original Research

BiomedRAG: A retrieval augmented large language model for biomedicine

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ABSTRACT

Retrieval-augmented generation (RAG) involves a solution by retrieving knowledge from an established database to enhance the performance of large language models (LLM). These models retrieve information at the sentence or paragraph level, potentially introducing noise and affecting the generation quality. To address these issues, we propose a novel BiomedRAG framework that directly feeds automatically retrieved chunk-based documents into the LLM. Our evaluation of BiomedRAG across four biomedical natural language processing tasks using eight datasets demonstrates that our proposed framework not only improves the performance by 9.95% on average, but also achieves state-of-the-art results, surpassing various baselines by 4.97%. BiomedRAG paves the way for more accurate and adaptable LLM applications in the biomedical domain.

1. Introduction

As research in the biomedical field deepens, breadth of biomedical literature technologies has increased. For instance, knowledge extraction technology aims to extract structured information from unstructured text [1,2]. Additionally, link prediction methods [3,4] focus on identifying the types of relationships between entities, which is beneficial for drug discovery [5]. To improve support for medical professionals and enhance the performance of BioNLP systems, Biomedical Large Language Models (LLMs) [6] provide a path through pre-training or fine-tuning open-source LLMs in the general domain using biomedical data. This demonstrates notable performance across a range of biomedical tasks. For instance, MedLLaMA [7] utilized biomedical literature for training and evaluation via biomedical question-answering tasks. GatorTronGPT [8] was trained on clinical texts and fine-tuned for various NLP tasks, such as biomedical relation extraction, biomedical question answering, link prediction, etc.

These models are commonly trained on extensive datasets, containing a significant amount of world or domain knowledge implicitly stored within their parameters. However, they are prone to hallucination [9,10]. Retrieval-augmented language models [11–14], in contrast,

can retrieve knowledge from an external datastore when needed, potentially reducing hallucination and increasing knowledge coverage. However, retrieving the entire sentence from the database introduces a lot of noise, in some noise-intensive tasks (where “noise-intensive” refers to the presence of irrelevant words in the input sentence that may impact label generation), input sentences are likely to introduce words irrelevant to the labels, and introducing noise that can adversely affect the performance of the model. For example, in the sentence-level triple extraction task, consider the sentence *Chitin synthetase was activated by fungal acid proteases; however, it was subsequently destroyed by proteases from both animal and plant sources*, the *STIMULATES* relationship between head entity *proteases* and tail entity *synthetase* is discerned from the key chunk *was activated by*. Moreover, previous methods involving retrieval-augmented language models typically require a fixed retriever, such as K-nearest neighbors (KNN) [15], to retrieve the most relevant document for the input sentence. However, it is unclear if the retrieved knowledge can adequately meet the LLM’s needs. Therefore, to solve the above challenges, we primarily investigate the effectiveness of integrating chunk knowledge into LLMs in the biomedical domain, and we propose a novel retrieval-augmented language framework for

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¹ Relation extraction: by giving the sentence and two entities in this sentence, the model needs to extract the relation between these two entities. Triple extraction: by giving the sentence, the model needs to joint extract the triple (head entity, relation, tail entity).

Table 1
Statement of Significance.

Problem or issue	What is already known	What this paper adds
Noise-intensive tasks (e.g., triple extraction, relation extraction) hinder LLM performance due to irrelevant information in input sentences.	Retrieval-augmented language models can reduce hallucinations but often retrieve entire sentences, introducing noise. Fixed retrievers like KNN may not meet model needs.	BiomedRAG introduces a tailored chunk scorer and retrieval-augmented framework to extract precise, relevant chunks. It significantly enhances performance in relation extraction, triple extraction, text classification, and link prediction tasks.

the biomedical domain, namely BiomedRAG, which is enhanced by a new tailored chunk scorer.

BiomedRAG consists of three major steps: (1) constructing the diverse chunk database. In our work, the “chunk” is a broad concept, For instance, in noise-intensive tasks, such as sentence-level relation extraction¹ and text classification tasks, the relation type or label typically pertains to several consecutive words in the sentence. Hence, the sentence is divided into multiple chunks. Conversely, in tasks like link prediction, where the input sentence contains only two entities and the output is the relation type, the chunk comprises these two entities. (2) training the tailored chunk scorer to select the relevant document from the diverse chunk database for the input sentence. (3) incorporating the retrieved document into the LLM to generate the output (e.g. label, structure knowledge, etc.) for the given sentence. We perform extensive experiments and demonstrate the effectiveness of our proposed BiomedRAG framework over four tasks (triple extraction, relation extraction, text classification, link prediction), showing significant improvement over strong baseline models (see Table 1).

2. Related work

2.1. Biomedical large language model

Recently, large language models (LLMs) like GPT-4,² and LLama3 [16] have become the dominant technology in various natural language processing (NLP) tasks, these models develop impressive capabilities in different NLP tasks, including information extraction (e.g., [12,17]), link prediction (e.g., [3,4,18]), and question answering (e.g., [19,20]). While much attention has been given to these models’ capabilities in the general domain, it is evident that specialist models have the potential to significantly help medical research [21]. To tailor LLMs for the medical domain, several specialized models have been developed, such as Bio-Mistral 7B [22], Medalpaca 13B [23] and CancerLLM [24]. However, LLM performance is highly sensitive to the selection of examples. The quality and relevance of examples provided during inference play a crucial role in shaping the model’s output. Poorly chosen, unclear, or irrelevant examples can lead to less accurate or even incorrect responses. To address the hallucination issue inherent in large language models (LLMs), our work explores a new retrieval-augmented method to improve the LLM performance in various medical tasks.

2.2. Retrieval-Augmented Language Models (RALMs)

Many studies [13,14,17,25–29], have used retrieved information from various knowledge stores to better interpret the text or generate the expected output. For example, KIRST [14] dynamically injects retrieved entity and attribute knowledge from the knowledge graph

when generating the entity or attribute in the task of entity stage changes. RAG [13] uses the maximum inner product search (MIPS) to find the top-K documents which are combined with a query to predict the answers. To enhance retrieval capability, RePLUG [29] treats the language model as a black box and enhances retrieval performance by leveraging the language model’s output. RT [30] utilizes KNN to retrieve the top-n documents from the database, which are then fed into the LLM to assist with information extraction. CTLP [3] proposes a Condensed Transition Graph Framework to retrieve paths between two entities and to predict the relationship between them. BIORAB [17] retrieves the most relevant document from the training set to improve the model performance in different biomedical tasks. Although these works achieve better performance than the LLM alone, they did not consider chunk-based knowledge. In this work, we investigate ways to improve language models by automatically retrieving chunk knowledge from a pre-constructed database.

3. Method

3.1. Datasets

3.1.1. Triple extraction dataset

In this paper, we utilized GIT, DDI, and Chemprot as the foundational datasets.

- GIT [31] is a high-quality biomedical triple extraction dataset for non-drug therapies, characterized by its high-quality annotations and comprehensive coverage of relation types. It includes 22 relation types from SemMedDB.
- CHEMPROT [32]: The Chemical Protein Interaction Corpus comprises 2432 PubMed abstracts annotated with chemical–protein interactions, encompassing 23 distinct interaction relations. Building upon prior research [33], the corpus exclusively considers sentence-level instances, with a particular focus on five prominent interaction types for classification: CPR3, CPR4, CPR5, CPR6, CPR9.
- DDI [34]: The DDI dataset was formulated to support drug information extraction (IE) research for SemEval 2013. It comprises 233 texts sourced from Medline abstracts and 792 texts from the DrugBank database. This dataset encompasses four distinct types of relations between drug entities, namely Advice, Mechanism, Effect, and Int.³

3.1.2. Relation extraction

In this paper, we utilized GIT-RE as the foundational dataset in relation extraction task. The biomedical triple extraction dataset GIT serves as the source data for the relation extraction task. We convert the GIT dataset into the relation extraction dataset GIT-RE by using the input context, head entity, and tail entity as model inputs, where the model’s output signifies the relationship between the input head entity and tail entity.

¹ Relation extraction: by giving the sentence and two entities in this sentence, the model needs to extract the relation between these two entities. Triple extraction: by giving the sentence, the model needs to joint extract the triple (head entity, relation, tail entity).

² <https://openai.com/index/gpt-4/>

³ This type is assigned when a sentence states that an interaction occurs and does not provide any information about the interaction.

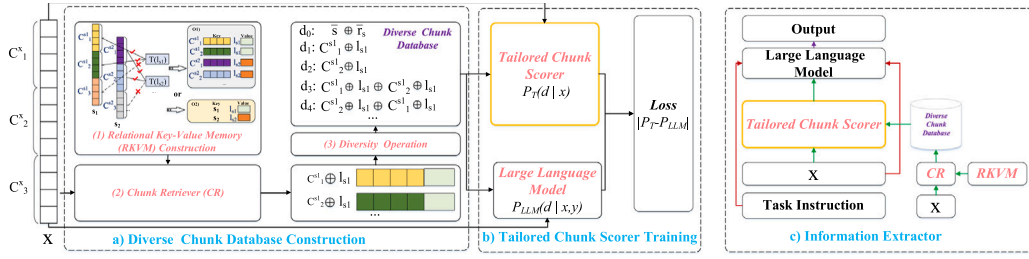


Fig. 1. Overview of BIOMEDRAG, which comprises three main steps: (a) construction of a diverse chunk database, (b) training of a tailored chunk scorer, and (c) information extraction. The scorer is trained to select the relevant information for the input sentence. Note: Chunk refers to several consecutive words in the input sentence. in our work, $\oplus$ refers to the combination of two sequences, for example, key="was activated by", value="STIMULATES", key $\oplus$ value="was activated by STIMULATES".

3.1.3. Text classification

In this paper, we utilized ade-corpus-v2 [35] and MTsample⁴ as the foundational dataset in the text classification task.

- ade-corpus-v2 dataset is designed for classifying whether a sentence is ADE(Adverse Drug Reaction)-related (True) or not (False). In our paper, we randomly selected 4000 instances for training, 500 for testing, and 500 for validation.
- The MTsample dataset is designed to analyze the nature of the language used in medical transcriptions of various kinds. It includes more than 40 classes.

3.1.4. Link prediction

In this paper, we utilized UMLS [36] and ADInt [37] as the foundational datasets in the link prediction task.

- UMLS [36] contains triples from the Unified Medical Language System, providing knowledge in the domain of healthcare and medicine. It comprises 6529 triples, divided into 5216 for training, 652 for validation, and 661 for testing.
- ADInt [37] is a dataset for identifying new pharmaceutical interventions (PI) for Alzheimer's Disease (AD). In our paper, we randomly selected 6000 samples from the source training set for training, 720 samples from the source testing set for testing, and 720 samples from the source validation set for validation.

3.2. OUR METHOD: BIOMEDRAG

Fig. 1 gives an overview our BIOMEDRAG, which consists of three major parts:

(a) *Diverse chunk database construction.* involves three substeps: (1) Develop the Relational Key-Value Memory (RKVM) M . (2) Employ the Chunk Scorer to retrieve the relevant key-value (chunk-label) pairs corresponding to the input sentence. (3) Construct the diverse chunk database through diversity operations, incorporating relevant key-value pairs along with the most pertinent (sentence, label) pair extracted from the validation dataset for each input sentence.

(b) *Tailored chunk scorer training.* The Tailored Chunk Scorer's training process centers on choosing the most relevant document from a diverse chunk database based on a given input sentence, with guidance from the LLM scores.

(c) *Information extractor.* The Information Extractor generates the output (e.g. relation type, structural knowledge, etc.) by combining the input with the document that has the highest weight score from the diverse chunk database.

3.2.1. Diverse chunk database construction

RKVM Construction: In this part, we introduce how to construct the key-value memory based on chunks to aid in biomedical applications. Specifically, we utilize the validation data to aid in constructing the source dataset $D = \{s, l_s\}$, where s denotes the sentence and l_s signifies the label of sentence s , for example, if this task is relation extraction, l_s refers the relation type of the entity pair in sentence s . Subsequently, the sentence s is divided into v chunks, each with a length of $m = w/v$, where w represents the length of the s . Following this, we compute the similarity between each label l_s and the v th chunk C_v^s in sentence s using the cosine similarity $S(\cdot)$, as follows:

$$\text{sim}(T(l_s), C_v^s) = S(\mathbf{E}(T(l_s)), \mathbf{E}(C_v^s))$$

where $T(l_s)$ is the text description of label l_s . $\mathbf{E}(\cdot)$ is the encoder function, we used MedLLaMA 13B [7] as $\mathbf{E}(\cdot)$ in our work.⁵ Subsequently, for each value l_s in M , we determine its associated key by selecting the top two chunks from the sentence s . RKVM M can be defined as:

$$M_r^s = \{(\underbrace{C_1^s}_{\text{key}}, \underbrace{l_s}_{\text{value}}), (\underbrace{C_2^s}_{\text{key}}, \underbrace{l_s}_{\text{value}})\}$$

$$M = \{M_r^s\}$$

where, C_1^s and C_2^s represent the top-1 and top-2 chunks, respectively, for the label l_s within the sentence s . For example, in Fig. 1, the sub memory $M_{l_1}^{s_1}$ pertaining to label l_1 regarding sentence s_1 can be defined as follows: $(C_1^{s_1}, l_{s_1})$ and $(C_2^{s_1}, l_{s_1})$. Note: In our work, for each dataset, only the validation data from the same set were used to build the chunk database.

Chunk Retriever: The chunk retriever aims to get the most relevant key-value (k - v) pairs from RKVM for the i th split chunk C_i^x which has the same length as C_v^s in input sentence x . More specifically, we use MedLLaMA 13b as our encoder to map each key k and chunk C_i^x to the embeddings $\mathbf{E}(k)$ and $\mathbf{E}(C_i^x)$. The similarity between the chunk embedding and key embedding is computed by the cosine similarity:

$$\text{sim}(k, C_i^x) = S(\mathbf{E}(k), \mathbf{E}(C_i^x))$$

Subsequently, the key with the highest similarity, along with its corresponding value, will serve as the retrieved key-value pair a_i for chunk C_i^x . So the retrieved key-value pairs A_x for the input sentence x can be represented by:

$$A_x = \{a_0, a_1, \dots, a_i\}, a_i = k_i \oplus v_i$$

⁴ <https://mtsamples.com/>

⁵ We first utilized the output of the last hidden state layer and applied the mean function to compute the sentence or chunk embedding.

3.2.2. Diversity operation

Retrieving diverse knowledge has shown the potential to enhance the generation capabilities in NLP tasks, as observed in tasks like Dialogue State Tracking [38]. In our study, we assume that diverse knowledge has the potential to offer more meaningful and rich information to guide the LLM and enhance its ability in expected output identification. Therefore, we use permutation as the diversity operation in our work. For example, if $A_x = \{a_0, a_1, a_3\}$, after the permutation function, it will become $A_x = \{a_0, a_1, a_3, a_0 \oplus a_1, a_2 \oplus a_3, a_0 \oplus a_1 \oplus a_3\}$. Specifically, the permutation operation is specifically applied to A_x . The resulting permutation of A_x , combined with the $(\bar{s}, \bar{l}_s)$ pair, constitutes the Diverse Chunk Database $\hat{A}_x$. Here, $(\bar{s}, \bar{l}_s)$ is selected from the source dataset $D = \{(s, l_s)\}$, demonstrating the highest cosine similarity with x .

$$\hat{A}_x = \{\underbrace{\bar{s} \oplus \bar{l}_s}_{d_0}, a_0, a_1, \dots, a_{i-1}, \underbrace{\oplus a_i}_{d_j}\}$$

3.2.3. Tailored chunk scorer training

Tailored Chunk Scorer: There exists a different weight value between input sentence x and document d_j in $\hat{A}_x$. Tailored Chunk Scorer aims to learn the weight value between input context x and each d_j . Specifically, the input sentence x and each document d_j in $\hat{A}_x$ are encoded into the sentence embedding $E(x)$ and document embedding $E(d_j)$. We then calculate the similarity of each document d_j by:

$$P_T(d_j|x) = \frac{e^{\text{sim}(x, d_j)/\eta}}{\sum_{d_j \in \hat{A}_x} e^{\text{sim}(x, d_j)/\eta}}$$

where η represents a hyperparameter that regulates the temperature of the softmax function. The document retrieval likelihood $P_T(d|x)$ is calculated by calculating the highest probability between document d_j and input sentence x .

Training the Tailored Chunk Scorer: We use the LLM as a scoring function to help train the Tailored Chunk Scorer and measure how much each document d_j could improve the LLM perplexity. In the training process, the d_j that makes the LLM's output as close as possible to the ground truth is considered to be providing the document that the LLM needs. Specifically, we first compute $P_{LLM}(y|d_j, x)$, the LLM probability of the ground truth y given the input sentence x and a document d_j . The higher the probability, the better the document d_j is at improving the LLM's perplexity. So we compute the LM likelihood of each document d_j as follows:

$$P_{LLM}(d|x, y) = \max(P_{LLM}(y|d_1, x), \dots, P_{LLM}(y|d_j, x))$$

The Tailored Chunk Scorer is trained by minimizing the loss function between the document retrieval likelihood and LM likelihood:

$$L = \frac{1}{|B|} \sum_{x \in B} |P_T(d|x) - P_{LLM}(d|x, y)|$$

3.2.4. Information extractor

We construct the instruction-based datasets for each biomedical application. Specifically, the dataset contains four components: (1) Instruction (I), a manually defined guide for the LLM to generate output (such as, triples) for each sentence. (2) Example, where we employ the trained tailored chunk scorer to assign weights to each document in $\hat{A}_x$ for each input context x . The document $\bar{d}_j$ with the highest weight score is considered as the example. (3) Input sentence x . (4) output t . The expected output t in our model is predicted by the function,

$$P(t|x) = P(t|I \oplus \bar{d}_j \oplus x)$$

In the training progress, instruction I , example $\bar{d}_j$, train input sentence x , with the ground answers are fed into the LLM to learn the mapping between the input and the correct answers. In the generation progress, instruction I , example $\bar{d}_j$, and test sentence x are fed into the LLM to generate the t of the x .

3.3. Baselines

To validate the effectiveness of our framework BIOMEDRAG, this section describes the baselines employed across various tasks. For the triple extraction, relation extraction, text classification, link prediction task, (1) **LLaMA family**, MedLLaMA 13B [7], LLaMA2 13b [39], Llama-3.1-8B, and (2) **GPT-4** are used. For the text classification and link prediction task, we also used **BERT** [40], **BioBERT** [41], **GatorTron** [42] as the baselines. Note: for all baselines, the GPT-4 model is in the zero shot setting, while all the other models are fine-tuned. In the next, we will introduce the individual models used in different tasks.

3.3.1. Triple extraction

We selected six open triple extraction models as the baseline for the triple extraction task, they are UniRel [43], OneRel [44], UIE (base) [2], UIE (large) [2], E2H (base) [45], and E2H (large) [45]. Since these models were not evaluated on our dataset, we fine-tuned them using the parameters provided by their respective papers. We also used Retrieval-augmented Large Language Model (RA-KNN- n). As shown in Fig. 3, we used the MedLLaMA 13B and LLaMA2 13B as the base model. To further assess the efficacy of BIOMEDRAG, we also employed two state-of-the-art biomedical triple extraction models JBUIM [46], and SPBERE [47] as baselines on the DDI and ChemProt datasets. Their respective results from their source papers. Due to the unavailability of the code, we were unable to report the results for GIT.

3.3.2. Relation extraction

(1) RT- n : We employ RT [12], a retrieval-augmented and chain-of-thoughts method, to extract the relation between the head and tail entity. n refers the top- n documents. (2) RA-KNN- n : it is consistent with the RA-KNN- n in the baseline of the triple extraction task. As shown in Fig. 3, we used the MedLLaMA 13B and LLaMA2 13B as the base model.

3.3.3. Text classification

(1) RA-KNN- n : As shown in Fig. 3, in Ade-corpus-v2, LLaMA2 13B serves as the base model, while in MTsample, GPT-4 serves as the base model.

3.3.4. Link prediction

(1) RA-KNN- n : As shown in Fig. 3, LLaMA2 13B serves as the base model in UMLS and ADInt.

3.4. Evaluation metrics

In the Triple Extraction task, same as [43,48], a triple is regarded as correct when its relation type, the head entity and the tail entity are all correct. For example, in the sentence: *Infusion of prostacyclin (PGI2) reportedly attenuates renal ischemic injury in the dog and the rat.*, triple *<Infusion, treats, rat>* is regarded as correct while *<injury, treats, rat>* is not. Following the evaluation method of the previous work [2,43–45], we evaluated all the models and reported the evaluation metric, including Micro Precision, Recall, and F1-score. For the relation extraction, text classification, and link prediction task, we follow the same evaluation metrics as triple extraction.

4. Results

In this section, we present our main results on BIOMEDRAG with a focus on several practical facets, including (1) Comparative evaluations of the BIOMEDRAG framework against other models across 4 tasks and 8 datasets. (2) Comparative assessments with the current RAG frameworks. (3) Module (tailored chunk scorer, diversity operation) assessment. (4) Framework performance under different chunk sizes.

Table 2

Results of various approaches on triple extraction, relation extraction over 4 datasets.

(1) Triple Extraction (TE)									(2) Relation Extraction (RE)				
TE Approach	DDI Precision	Recall	F1	ChemProt Precision	Recall	F1	GIT Precision	Recall	F1	RE Approach	GIT-RE Precision	Recall	F1
UniRel [43]	25.59	21.92	27.13	29.00	17.45	21.79	36.36	12.50	18.60	RT-10 [12]	42.80	43.44	43.12
OneRel [44]	34.72	81.07	22.09	44.95	45.22	44.68	47.97	52.01	44.52	RT-5 [12]	44.91	46.45	45.67
UIE (base) [2]	30.74	29.07	29.88	48.15	44.79	46.41	34.81	30.59	32.56	RT-1 [12]	44.86	46.02	45.44
E2H (large) [45]	30.24	30.16	30.20	49.86	47.78	48.80	31.47	26.89	29.00	RT-20 [12]	44.94	45.81	45.37
E2H (base) [45]	34.23	34.25	34.24	48.92	46.51	47.68	31.29	26.68	28.80	BERT [40]	86.25	86.25	86.25
UIE (large) [2]	37.50	35.05	36.24	49.74	46.27	47.94	33.99	29.93	31.83	BioBERT [41]	85.43	85.43	85.43
GPT-4 [49]	16.27	20.65	18.20	20.79	37.00	26.61	9.48	9.46	9.47	GPT-4 [49]	47.09	47.09	47.09
MedLLaMA 13B [7]	76.63	76.63	76.63	52.10	49.04	50.52	42.60	41.51	42.05	MedLLaMA 13B [7]	67.02	66.88	66.95
LLaMA2 13B [39]	79.61	79.61	79.61	78.58	76.41	77.48	61.76	56.45	58.99	LLaMA2 13B [39]	79.43	78.92	79.18
LLaMA-3.1-8B	65.21	65.21	65.21	63.38	55.23	59.03	62.58	60.43	61.48	LLaMA-3.1-8B	81.16	80.64	80.90
MedLLaMA 13B + BIOMEDRAG	76.90	76.90	76.90	77.48	76.46	76.97	76.88	76.56	76.72	MedLLaMA 13B + BIOMEDRAG	86.66	86.66	86.66
LLaMA2 13B + BIOMEDRAG	80.50	79.10	80.00	89.25	88.42	88.83	81.78	81.07	81.42	LLaMA2 13B + BIOMEDRAG	89.03	89.03	89.03
GPT 4 + BIOMEDRAG	18.04	26.08	21.33	33.74	31.58	32.45	18.04	26.08	21.33	GPT4 + BIOMEDRAG	48.60	48.60	48.60
LLaMA-3.1-8B + BIOMEDRAG	73.36	73.36	73.36	75.57	70.58	72.99	62.72	60.43	61.55	LLaMA-3.1-8B + BIOMEDRAG	81.50	81.50	81.50

Table 3

Results of various approaches on text classification, link prediction over 4 datasets.

(1) Text classification						(2) Link prediction						
Approach	Ade-corpus-v2			MTsample			UMLS			ADInt		
	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
BERT [40]	95.00	97.00	96.00	38.00	38.00	38.00	57.00	57.00	57.00	58.15	58.15	58.15
BioBERT [41]	95.00	97.00	97.00	37.00	37.00	37.00	58.00	58.00	58.00	59.37	59.37	59.37
GatorTron [42]	95.00	98.00	97.00	38.00	38.00	38.00	59.00	59.00	59.00	59.23	59.23	59.23
GPT-4 [49]	41.00	41.00	41.00	41.33	41.33	41.33	6.00	6.00	6.00	33.33	33.33	33.33
MedLLaMA 13B [7]	95.40	95.40	95.40	24.27	23.14	23.69	34.51	32.37	33.41	46.25	46.25	46.25
LLaMA2 13B [39]	96.40	96.40	96.40	38.49	36.80	37.62	56.12	56.12	56.12	61.38	61.38	61.38
LLaMA-3.1-8B	91.16	66.00	76.56	34.80	34.80	34.80	56.88	56.88	56.88	55.83	55.83	55.83
GPT-4 + BIOMEDRAG	75.60	75.60	75.60	41.93	41.93	41.93	24.35	24.35	24.35	37.08	37.08	37.08
MedLLaMA 13B + BIOMEDRAG	99.89	99.89	99.89	32.52	32.45	32.49	58.00	58.00	58.00	59.16	59.16	59.16
LLaMA2 13B + BIOMEDRAG	99.80	99.80	99.80	38.50	38.50	38.50	59.80	59.80	59.80	62.22	62.22	62.22
LLaMA-3.1-8B + BIOMEDRAG	99.80	98.40	99.19	42.27	41.93	42.10	57.49	57.49	57.49	56.66	56.66	56.66

Table 4

Triple, head entity, tail entity and relation results of various approaches on DDI, ChemProt and GIT.

	Approach	Triple			Head entity			Tail entity			Relation		
		Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
DDI	UIE (large)	37.50	35.05	36.24	68.89	64.40	66.57	63.37	59.24	61.23	45.63	42.66	44.10
	LLaMA2 13B + BIOMEDRAG	80.50	79.10	80.00	97.01	97.01	97.01	85.59	85.59	85.59	92.11	92.10	92.12
ChemProt	E2H (large)	49.86	47.78	48.80	80.45	77.09	78.73	78.23	74.97	76.56	71.26	68.29	69.74
	LLaMA2 13B + BIOMEDRAG	89.25	88.42	88.83	98.26	97.35	97.80	96.71	95.81	96.25	93.01	92.14	92.57
GIT	OneRel	47.97	52.01	44.52	65.26	71.54	60.00	68.16	75.46	62.15	74.70	82.94	67.96
	LLaMA2 13B + BIOMEDRAG	81.78	81.07	81.42	92.84	92.04	92.44	91.76	90.96	91.36	87.20	86.45	86.82

4.1. Comparative assessments between BIOMEDRAG framework with other models

Table 2 and Table 3 present the experiment results on triple extraction, relation extraction, text classification, and link prediction. Through our experiments, we noted that our BIOMEDRAG model exhibited the capacity to enhance the performance of different LLMs such as LLaMA2 13B, MedLLaMA 13B, LLaMA3.1 8B and GPT-4. LLaMA2 13B, MedLLaMA 13B, and LLaMA3.1 8B are all fine-tuned, while GPT-4 is used in a zero-shot setting.

Triple extraction. In our paper, triple extraction is defined as the task where the model extracts the triple (head entity, relation, tail entity) from sentences.

From Table 2 (1), we have the following observations: On GIT, (1) our BIOMEDRAG significantly outperforms all the strong baselines across all evaluation metrics. (2) we observe that BIOMEDRAG improved the original MedLLaMA 13B, LLaMA2 13B, GPT-4 by 34.67%, 22.43%, 11.86% respectively, in terms of Triple-F1. (3) The performance of lightweight models like UniRel and OneRel significantly lags behind

that of the LLaMA2 family. On DDI and ChemProt, we have the following observations: (1) our BIOMEDRAG still outperforms all baselines across F1 value on these open datasets. (2) We observe that BIOMEDRAG improved the original MedLLaMA 13B, LLaMA2 13B, LLaMA-3.1-8B and GPT-4 by 0.27%, 0.39%, 8.51%, 3.13% respectively on the DDI. (3) We observe that BIOMEDRAG improved the original MedLLaMA 13B, LLaMA2 13B, LLaMA-3.1-8B and GPT-4 by 26.45%, 11.35%, 13.96%, 5.84% respectively on the ChemProt. (4) The lower performance observed in UniRel and OneRel can be attributed to the table-filling method⁶ struggling to recognize complex biomedical entities or relations. For example, as illustrated in Table 4, on DDI, UIE achieves only a 61.23% accuracy for tail entity recognition.

Table 4 presents the entities, relation types, and triple comparisons among the top-1 baseline models from Table 2 and BIOMEDRAG. We

⁶ The table-filling method for triple extraction involves parsing sentences to identify entities (subjects and objects) and relationships. These components are used to form triples (subject, predicate, object), which are then organized into a table, with each row representing a relationship. Multiple relationships in a sentence result in multiple rows.

observed that (1) BIOMEDRAG achieved state-of-the-art performance in triple F1, relation F1 and entity F1. (2) For the lightweight model, such as UIE, it achieves a lower F1 score for relations on the DDI dataset. (3) Lightweight models like UIE and OneRel struggle with entity recognition in biomedical sentences. To further validate the performance of our model, we present the results of triple extraction from several state-of-the-art systems: SPBERE [47] and JBUIM [46], as sourced from their respective papers. Since their code is not publicly available, we compare their model performance solely on the DDI and ChemProt datasets. Our findings indicate that our model continues to demonstrate superior performance compared to these systems. On the DDI dataset, JBUIM and SPBERE achieved F1 scores of 77.70% and 79.20% respectively, whereas our model achieved an F1 score of 80.00%. Similarly, on the ChemProt dataset, JBUIM and SPBERE achieved F1 scores of 68.80% and 69.70%, while our model achieved an F1 score of 88.83%.

Relation extraction. In our paper, the relation extraction task involves predicting the type of relationship between two entities (head entity and tail entity) based on a given sentence, head entity, and tail entity. To assess the scalability of our model, in this part, we evaluate the performance of BIOMEDRAG in relation extraction tasks. Table 2 (2) presents the experiment results of various approaches. We have the following observations: (1) our BIOMEDRAG significantly outperforms all the strong baselines and its variants across all evaluation metrics. (2) We observed that BIOMEDRAG improved the original MedLLaMA 13B, LLaMA2 13B, LLaMA-3.1-8B, and GPT-4 by 19.71%, 9.85%, 0.60%, and 1.51% respectively, in term of F1. (3) Without training, GPT-4 struggles to extract the relationships in the sentence.

Text classification. Table 3 (1) presents the experiment results of various approaches on the two datasets ade-corpus-v2 and MT-sample. On Ade-corpus-v2, (1) It is notable that BIOMEDRAG enhances the original MedLLaMA 13B, LLaMA2 13B, LLaMA-3.1-8B and GPT-4 by 4.49%, 3.40%, 22.63% and 34.60% respectively, in terms of F1 score. (2) Without fine-tuning, GPT-4 exhibits lower performance on the ade-corpus-v2 dataset. Nevertheless, BIOMEDRAG has the capability to enhance the performance of GPT-4, surpassing the performance achieved by BERT, BioBERT, and GatorTron. On MT-Sample, we observed that: (1) GPT-4, without fine-tuning, achieves the best performance when compared to models like LLaMA2 13B and MedLLaMA 13B, which require fine-tuning. (2) Remarkably, BIOMEDRAG boosts the performance of the original MedLLaMA 13B, LLaMA2 13B, LLaMA-3.1-8B, and GPT-4 models by 8.80%, 0.88%, 7.30 and 0.60%, respectively, in F1 score.

Link prediction. Table 3 (2) presents the experiment results of various approaches on the UMLS and ADInt. On UMLS, we can see that: (1) BIOMEDRAG improved the original MedLLaMA 13B, LLaMA2 13B, LLaMA-3.1-8B and GPT-4 by 24.59%, 3.68%, 0.61% and 18.35% respectively, in term of F1. On ADInt, (1) we observed that BIOMEDRAG improved the original MedLLaMA 13B, LLaMA2 13B, LLaMA-3.1-8B and GPT-4 by 12.91%, 0.84%, 0.83% and 3.75% respectively, in term of F1.

4.2. Comparative assessments with other RAG frameworks

The core of our framework design lies in establishing a relational key-value memory at the chunk level and training a tailored chunk scorer to adapt the LLM. To comprehensively assess our model's performance, we conduct a comparative analysis with the prevailing retrieval-based LLM, employing a retriever to obtain the top- n relevant documents related to the input sentence. we called this model as RA-KNN- n . More specifically, same as [12,28], we employ K-nearest neighbors (KNN) as the retriever to obtain the top- n (sentence, label) pairs from the retrieval database (same with the retrieval database as our framework). The model results of different n are shown in Fig. 3.

In the experiment of GIT and GIT-RE, we selected LLaMA2 13B and MedLLaMA 13B as LLM. Due to input length constraints, we present

the performance results for the top-15 in MedLLaMA 13B and top-30 in LLaMA2 13B. As shown in Fig. 2(c), upon comparing the F1 values between OneRel and MedLLaMA 13B+top1, we observe that the retrieval-based model proves effective in enhancing triple extraction performance. As shown in Fig. 2 (c, d), with the incorporation of BIOMEDRAG, we observed an enhancement in the performance of both MedLLaMA 13B and LLaMA2 13B beyond the best achieved through retrieval alone. Otherwise, we observed that even though our model retrieves only the top-1 document from the diverse chunk database, it still outperforms the KNN-based RAG-LLM in retrieving the top-20 documents. For instance, in Fig. 2(d), BIOMEDRAG achieves 89.03%, whereas LLaMA2 13B+top20 only achieves 88.38% on GIT. It is also crucial to highlight that while retrieving more sentences can improve the performance of the original LLM, it also leads to increased training time. Despite the superior performance of LLaMA2 13B+top15 compared to LLaMA2 13B+BR, as demonstrated in Fig. 2(h), retrieving larger documents will impact both training and inference times, along with consuming additional computational resources.

From Fig. 3(a, b, c, d, e, f, g, h), We have discovered that retrieving as much as documents is not necessary to enhance the model's performance, for example, Fig. 3(b), as when $n = 30$, the model gets a worse performance. For both the triple extraction (Fig. 3(c)), text classification task (Fig. 3(e)) and relation extraction tasks (Fig. 3(c)), we observed that the model achieves its best performance when $n = 1$. Retrieving documents appears to be beneficial for enhancing label prediction accuracy. In Fig. 3, we observe that not all tasks benefit from using the LLM with the KNN-based retrieval method. For instance, on the DDI dataset, the RA-KNN- n model exhibits lower performance compared to the LLM without example-guided generation. However, our BIOMEDRAG demonstrates superior performance when compared to both the RA-KNN- n and other base models.

4.3. Model performance under different chunk sizes

In our proposed model, there is a parameter that controls the number of chunks: $m \in \{3, 4, 5\}$ when constructing the relational key-value memory, which represents the length of each chunk. This parameter aids in retrieving relevant chunk information, considering that the input sentence often contains noise that can impact generation progress. In this section, we show the model performance on two tasks triple extraction and relation extraction—that have shown significant improvement by BIOMEDRAG, as shown in Fig. 4(a–c) and Fig. 4(d). As we can see, in these two tasks, when $m = 5$, BIOMEDRAG can construct the more rich relation-based information, and retrieve the more relevant relation chunk to the input sentence. In addition, the performance of $m = 5$ is better than $m = 3$ or $m = 4$, also indicating that if the granularity of relation chunk is too coarse, it will impact relation recognition and consequently affect model performance.

4.4. Module (tailored chunk scorer, diversity operation) assessment

The contribution of our model components can also be learned from ablated models. In this part, we mainly focus on biomedical triple extraction and relation extraction task, which get significant improvement over our BIOMEDRAG. We introduce two ablated models of BIOMEDRAG, (1) BIOMEDRAG-WTCS uses a cosine similarity to choose the most relevant document from diverse chunk database, without using the Tailored Chunk Scorer. (2) BIOMEDRAG-WD does not consider the diversity of the chunk database. In these experiments, we selected LLaMA2 13B + BIOMEDRAG as it demonstrated the highest performance from Table 2. We find that the performance of BIOMEDRAG degrades as we remove important model components. Specifically, as shown in Fig. 5 both BIOMEDRAG-WTCS and BIOMEDRAG-WD performs poorly when compared to BIOMEDRAG, indicating the importance of training a tailored chunk scorer to adapt the LLM and improve the diversity of chunk database.

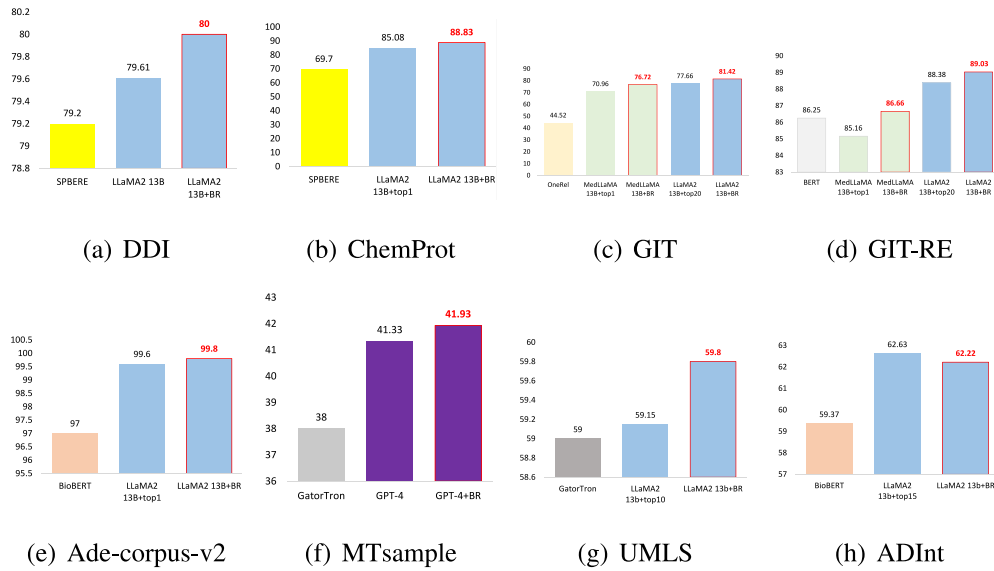


Fig. 2. F1 performance of different models. BR refers to BIOMEDRAG. The red font indicates the performance of BIOMEDRAG.

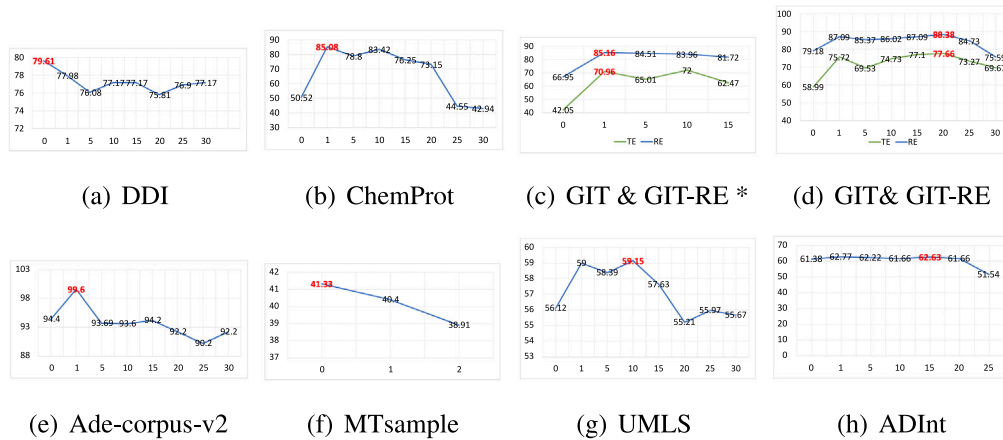


Fig. 3. Different F1 (a-h) performance of KNN-based retrieval model on 4 tasks over 8 datasets. We select the LLM with the highest baseline performance as our primary LLM model. The number of document samples depends on the input length constraint of the LLM. For instance, in dataset Ms-sample, when $top-n > 2$, the maximum input token length of 8318 exceeds the maximum input token length of 8192 tokens for GPT-4. y-axis: F1 (a-h). x-axis: 0 represents no retrieval, while 1–30 represents the top-n documents retrieved. The red font refers to the best performance. The difference between (c) and (d) lies in the LLM used: in (c), the LLM is MedLLaMA 13B, while in (d), the LLM is LLaMA2 13B.

5. Discussion

In this paper, we introduce BIOMEDRAG, a novel framework aimed at enhancing the performance of original Large Language Models (LLMs) such as MedLLaMA 13B, LLaMA2 13B. Our proposed framework leverages the retrieved chunk documents, acquired through a specifically trained tailored chunk scorer, to augment the capabilities of LLMs. This improvement highlights the efficacy of integrating retrieved chunk documents into the LLM framework. For example, by comparing our model with the current RAG model as shown in Fig. 2, we observe that even when retrieving only the top-1 document from the diverse chunk database, our model outperforms the KNN-based RAG-LLM in retrieving the top-n documents. This suggests that BIOMEDRAG excels in retrieving diverse chunk documents by utilizing a tailored chunk scorer, thereby significantly enhancing the overall performance of the model.

On the triple extraction task, GPT-4 demonstrates notably lower performance, particularly in the case of GIT. The main reason is that the reported results are in the zero-shot setting due to the unavailability of open resources. UniRel performs significantly worse than OneRel,

primarily due to the superiority of the BIE operation in OneRel compared to the Interaction Map in UniRel for entity recognition. However, the lower performance observed in UniRel for entity recognition. However, the lower performance observed in UniRel can be attributed to the table-filling method struggling to recognize complex biomedical entities or relations. For instance, OneRel achieves only 62.15% for tail entity recognition on GIT. It remains challenging to recognize some complex biomedical words. In the realm of triple extraction tasks, GPT-4 exhibits markedly diminished performance, particularly evident in its performance on GIT. This decline can be chiefly attributed to the fact that the reported outcomes stem from a zero-shot setting, necessitated by the absence of accessible resources. UniRel finds itself notably outpaced by OneRel, primarily owing to the superior efficacy of the BIE operation within OneRel in contrast to the Interaction Map employed in UniRel for entity recognition. Nonetheless, the subdued performance witnessed in both UniRel and OneRel can be ascribed to the limitations of the table-filling method in accurately identifying intricate biomedical entities or relationships. For instance, OneRel achieves a meager 62.15% accuracy for tail entity recognition on GIT, underscoring the persistent challenge of accurately recognizing certain complex biomedical terms.

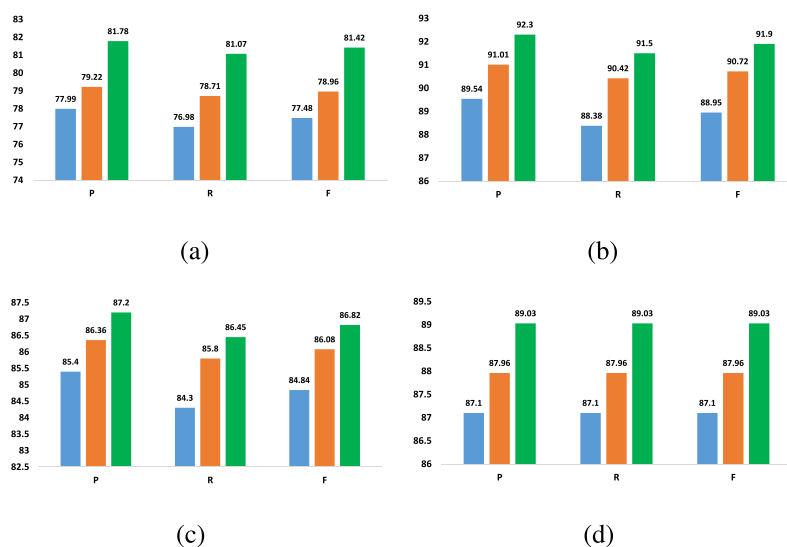


Fig. 4. (a–c): P, R, F1 results with different chunk length m settings in the task of triple extraction task, (a: triple performance, b: entity performance, c: relation performance). (d): P, R, F1 results with different chunk length m settings in the task of relation extraction task. The blue, orange, and green bars represent the performance of BiomedRAG with settings of $m = 3, 4, 5$.

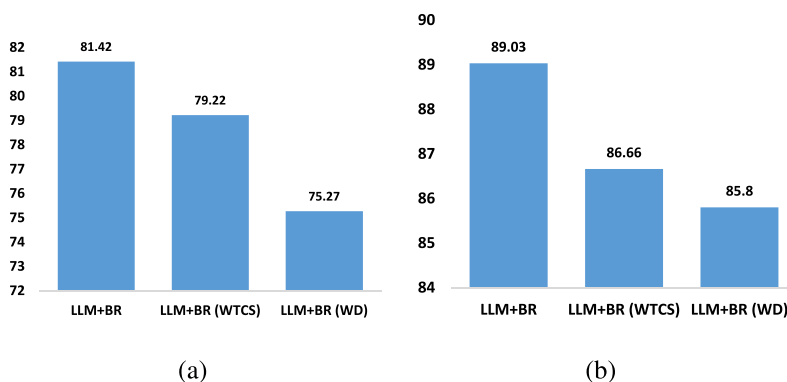


Fig. 5. BIOMEDRAG performance about its ablated model. we chose the LLaMA2 13B as the LLM. (a): triple extraction. (b): relation extraction. BR: **biomedRAG**.

It is noteworthy that during experimentation on both GIT and GIT-RE, we observed intriguing results when varying the chunk number parameter. Specifically, when setting $m = 5$, BIOMEDRAG demonstrated an enhanced capacity to construct richer relation-based information and retrieve more relevant relation chunks corresponding to the input sentence. Moreover, the performance achieved with $m = 5$ surpassed that of $m = 3$ or $m = 4$, highlighting the detrimental impact of overly coarse granularity on relation recognition and subsequently on model performance. Furthermore, the efficacy of this particular chunk number extends beyond the GIT dataset, proving beneficial in tackling noise-intensive tasks and datasets such as ChemProt, DDI, and Ade-corpus-v2. This underscores the versatility and effectiveness of utilizing an optimal chunk number parameter in enhancing model performance across various domains and challenging datasets. Otherwise, our exploration of the MTsample dataset revealed a fascinating phenomenon: even without fine-tuning, GPT-4 exhibited superior performance compared to the larger language model such as LLaMA2 13B, even when fine-tuned. We postulate that GPT-4's ability to harness its inherent knowledge for performance enhancement during prompt tuning contributes to this phenomenon.

6. Limitation

One limitation of our approach is the design of instructions for optimizing GPT-4's performance. In future work, we plan to explore

various instruction designs to enhance generation quality. In this paper, we focused solely on using chunk information; in the future, we intend to use grammatical units rather than simple word chunks to better preserve semantic relationships within each segment. Additionally, we used the validation set to construct the retrieval dataset, which may impact generalization. To address this, we aim to adopt a human-guided method for building the retrieval dataset in future work.

7. Conclusion

We introduce a novel biomedical RAG framework called BIOMEDRAG. Unlike the traditional retrieval-argument language model, our framework retrieval the knowledge from the diverse chunk database and adapts the tailored chunk scorer to the LLM. Experimental results show that our framework achieves consistent improvements across 4 biomedical NLP tasks over 8 datasets. Various LLM with BiomedRAG framework achieved the better performance on these tasks. BiomedRAG has demonstrated a great value to biomedicine NLP field.

CRedit authorship contribution statement

Mingchen Li: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Data curation, Conceptualization. **Halil Kilicoglu:** Writing – review & editing. **Hua Xu:** Writing – review & editing. **Rui Zhang:** Writing – review & editing, Writing – original

draft, Supervision, Resources, Methodology, Funding acquisition.

Code Availability

The complete code will be available in the repository: <https://github.com/ToneLi/BioMedRAG/tree/main>

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Prompting examples for fine-tuning across different tasks

A.1. Prompting example in triple extraction (GIT dataset)

“instruction”: “please extract the triplet from the given sentence, the triplet is [head entity, relation, tail entity], the element relation denotes the relationship between head entity and tail entity. the relation must in my predefined relation set: ('AFFECTS', 'INTERACTS WITH', 'TREATS', 'PRODUCES', 'PRECEDES', 'PREDISPOSES', 'ADMINISTERED TO', 'PROCESS OF', 'AUGMENTS', 'PREVENTS', 'DIAGNOSES', 'COEXISTS WITH', 'ASSOCIATED WITH', 'DISRUPTS', 'CAUSES', 'COMPLICATES', 'SYMPTOM OF', 'DOES NOT TREAT', 'STIMULATES', 'MANIFESTATION OF', 'USES'). response Format: head entity|relation|tail entity”, **“context”**: “[on the effect of respiration of a combination of morphine -like acting pethidine with the morphine antagonist nalorphine]”, **“response”**: “nalorphine|INTERACTS WITH|morphine”, **“category”**: “triplet extraction”

A.2. Prompting example in relation extraction (GIT-RE dataset)

“instruction”: “You are an excellent medical expert. The task is to predict the relationship between the given head entity and tail entity within a given sentence, this relation which must be in ('PRECEDES', 'PREDISPOSES', 'TREATS', 'AFFECTS', 'INTERACTS WITH', 'INHIBITS', 'PRODUCES', 'ADMINISTERED TO', 'PROCESS OF', 'AUGMENTS', 'PREVENTS', 'DIAGNOSES', 'COEXISTS WITH', 'ASSOCIATED WITH', 'DISRUPTS', 'CAUSES', 'COMPLICATES', 'SYMPTOM OF', 'DOES NOT TREAT', 'STIMULATES', 'MANIFESTATION OF', 'USES')”, **“context”**: “Input: In sentence natural history of hiv infection in european drug users . what is the relationship between hiv infection and drug users?”, **“response”**: “PROCESS_OF”, **“category”**: “relation_extraction”

A.3. Prompting example in text classification (Ade dataset)

“instruction”: “You are an excellent medical expert. The task is to predict label about this sentence, the label is true or false for the given sentence”, **“context”**: “We conjectured that the side effects of insulin, such as anti-natriuresis and increased vascular permeability, might be pronounced in the presence of the hepatic dysfunction that accompanies insulin insensitivity, hyperinsulinaemia and hypoalbuminaemia”, **“response”**: “False”, **“category”**: “classification”

A.4. Prompting example in link prediction (ADInt dataset)

“instruction”: You are an excellent medical expert. The task is to predict the relationship between two entities, the relationship must be in the set ('uses', 'affects', 'prevents', 'indicates', 'conceptual part of', 'measurement of', 'interacts with', 'analyzes', 'ingredient of', 'method of', 'degree of', 'assesses effect of', 'result of', 'exhibits', 'manifestation of', 'adjacent to', 'performs precedes', 'disrupts', 'cooccurs with', 'part of', 'associated with', 'isa', 'produces', 'evaluation of', 'issue in', 'carries out', 'causes', 'complicates', 'treats', 'diagnoses', 'measures', 'location of', 'occurs in', 'process of', 'property of') **“context”**: “what is the relationship between human caused phenomenon or process and pathologic function ?” **“response”**: “result of”, **“category”**: “link prediction”

A.5. Parameters of fine tuning

per device train batch size=1, gradient accumulation steps=4, gradient checkpointing=True, logging steps=1, learning rate=2e-4, num train epochs=5, max grad norm=0.3, warmup ratio=0.03, output dir = “llama3 8b”, optim=“paged adamw 8bit”, fp16=True, evaluation strategy=“epoch”, eval steps=0.2, metric for best model=“loss”, greater is better=False, save strategy=“epoch”, save total limit=1, max seq length=4000

Appendix B. Prompting examples for zero-shot across different tasks (GPT-4)

B.1. Prompting example in triple extraction (Chemprot)

SYSTEM PROMPT = “Please extract the triplet from the sentence, which consists of [head entity, relation, tail entity]. The relation element indicates the relationship between the head entity and the tail entity, and it must conform to our defined relations” USER PROMPT = “Are you clear about your role?” ASSISTANT PROMPT = “Sure, I'm ready to help you with your triplet extraction task. Please provide me with the necessary information to get started”. GUIDELINES PROMPT = (“relations: PRODUCT-OF, DOWNREGULATO, ANTAGONIST,AGONIST, ACTIVATOR” “Sentence: ” “Output: ”

B.2. Prompting example in relation extraction (GIT-RE)

SYSTEM PROMPT = “Please identify the relationship between the two entities in the sentence by considering the provided candidate relations. In other words, select the relation from the available options that best aligns with the sentence regarding the two given entities”. USER PROMPT 1 = “Are you clear about your role?” ASSISTANT PROMPT 1 = “Sure, I'm ready to help you with your relation extraction task. Please provide me with the necessary information to get started”. GUIDELINES PROMPT = (“relation Definition:” “1. CAUSES: Brings about a condition or an effect. Implied here is that an agent, such as for example, a pharmacologic substance or an organism, has brought about the effect. This includes induces, effects, evokes, and etiology. Neurocysticercosis (NCC) is one of the major causes of neurological disease ... ” “2.COMPLICATES: Causes to become more severe or complex, or results in adverse effects “3.USES: Employs in the carrying out of some activity. This includes applies, utilizes, employs, and avails. ” “4.STIMULATES: Increases or facilitates the action or function of (substance interaction). ” “5. DISRUPTS: Alters or influences an already existing condition, state, or situation. Produces a negative effect on. ” “6. TREATS: Applies a remedy with the object of effecting a cure or managing a condition ” “7. COEXISTS WITH: Occurs together with, or jointly. Food intolerance-related constipation is characterized by proctitis. ” “8. MANIFESTATION OF: That part of a phenomenon which is directly observable or concretely or visibly expressed, or which gives evidence to the underlying process. This includes expression of, display

of, and exhibition of. ” “9. INTERACTS WITH: Substance interaction. ” “10. ADMINISTERED TO: Given to an entity, when no assertion is made that the substance or procedure is being given as treatment. ” “11. PREVENTS: Stops, hinders or eliminates an action or condition ” “12. PREDISPOSES: To be a risk to a disorder, pathology, or condition ” “13. INHIBITS: Decreases, limits, or blocks the action or function of (substance interaction). ” “14. AUGMENTS: Expands or stimulates a process. ” “15. PRODUCES: Brings forth, generates or creates. This includes yields, secretes, emits, biosynthesizes, generates, releases, discharges, and creates ” “16. PROCESS OF: Disorder occurs in (higher) organism. ” “17. PRECEDES: Occurs earlier in time. This includes antedates, comes before, is in advance of, predates, and is prior to. ” “18. AFFECTS: Produces a direct effect on. Implied here is the altering or influencing of an existing condition, state, situation, or entity. This includes has a role in, alters, influences, predisposes, catalyzes, stimulates, regulates, depresses, impedes, enhances, contributes to, leads to, and modifies. ” “19. DIAGNOSES: Distinguishes or identifies the nature or characteristics of ” “20. ASSOCIATED WITH: Has a relationship to (genedisease relation)”. “21. DOES NOT TREAT: antonyms of TREATS” “22. SYMPTOM OF: departure from normal function or feeling which is noticed by a patient, reflecting the presence of an unusual state, or of a disease; subjective, observed by the patient, cannot be measured directly”

B.3. Prompting example in text classification(ADE)

SYSTEM PROMPT = f“You are an excellent linguist. The task is for Classification if a sentence is ADE-related (True) or not (False)” USER PROMPT 1 = “Are you clear about your role?” ASSISTANT PROMPT 1 = “Sure, I’m ready to help you with your classification task task. Please provide me with the necessary information to get started”.

B.4. Prompting example in link predictions (UMLS)

SYSTEM PROMPT = “You are an excellent linguist. The task is to predict the relationship between two entities, the relationship must be in the set ‘uses’, ‘affects’, ‘prevents’, ‘indicates’, ‘conceptual part of’, ‘measurement of’, ‘interacts with’, ‘analyzes’, ‘ingredient of’, ‘method of’, ‘degree of’, ‘assesses effect of’, ‘result of’, ‘exhibits’, ‘adjacent to’, ‘performs’, ‘precedes’, ‘disrupts’, ‘manifestation of’, ‘part of’, ‘associated with’, ‘isa’, ‘produces’, ‘co occurs with’, ‘evaluation of’, ‘treats’, ‘carries out’, ‘causes’, ‘issue in’, ‘complicates’, ‘occurs in’, ‘process of’, ‘property of’, ‘diagnoses’, ‘measures’, ‘location of’” USER PROMPT 1 = “Are you clear about your role?” ASSISTANT PROMPT 1 = “Sure, I’m ready to help you with your link prediction tasks. Please provide me with the necessary information to get started”.

Data availability

All the datasets involved in this study are publicly available from the following official websites: CHEMPROT: <https://biocreative.bioinformatics.udel.edu/news/corpora/chemprot-corpus-biocreative-vi/DDI:https://aclanthology.org/S13-2056/> GIT: https://github.com/ToneLi/BioMedRAG/tree/main/dataset/0_GM-CIHT, <https://arxiv.org/pdf/2310.18463ade-corpus-v2>: https://huggingface.co/datasets/ade-benchmark-corpus/ade_corpus_v2MTsample: <https://mtsamples.com/ADInt:https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10246059/UMLS:https://www.nlm.nih.gov/research/umls/quickstart.html>.

References

- [1] L. Luo, Z. Yang, M. Cao, L. Wang, Y. Zhang, H. Lin, A neural network-based joint learning approach for biomedical entity and relation extraction from biomedical literature, *J. Biomed. Inform.* 103 (2020) 103384.
- [2] Y. Lu, Q. Liu, D. Dai, X. Xiao, H. Lin, X. Han, L. Sun, H. Wu, Unified structure generation for universal information extraction, 2022, arXiv preprint [arXiv:2203.12277](https://arxiv.org/abs/2203.12277).
- [3] M. Li, C. Ling, R. Zhang, L. Zhao, A condensed transition graph framework for zero-shot link prediction with large language models, 2024, arXiv preprint [arXiv:2402.10779](https://arxiv.org/abs/2402.10779).
- [4] M. Li, J. Chen, S. Mensah, N. Aletras, X. Yang, Y. Ye, A hierarchical n-gram framework for zero-shot link prediction, 2022, arXiv preprint [arXiv:2204.10293](https://arxiv.org/abs/2204.10293).
- [5] R. Zhang, D. Hristovski, D. Schutte, A. Kastrin, M. Fiszman, H. Kilicoglu, Drug repurposing for COVID-19 via knowledge graph completion, *J. Biomed. Inform.* 115 (2021) 103696.
- [6] C. Ling, X. Zhao, J. Lu, C. Deng, C. Zheng, J. Wang, T. Chowdhury, Y. Li, H. Cui, T. Zhao, et al., Domain specialization as the key to make large language models disruptive: A comprehensive survey, 2305, 2023, *ArXiv Preprint ArXiv*.
- [7] C. Wu, X. Zhang, Y. Zhang, Y. Wang, W. Xie, Pmc-llama: Further finetuning llama on medical papers, 2023, arXiv preprint [arXiv:2304.14454](https://arxiv.org/abs/2304.14454).
- [8] C. Peng, X. Yang, A. Chen, K.E. Smith, N. PourNejatian, A.B. Costa, C. Martin, M.G. Flores, Y. Zhang, T. Magoc, et al., A study of generative large language model for medical research and healthcare, *NPJ Digit. Med.* 6 (1) (2023) 210.
- [9] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y.J. Bang, A. Madotto, P. Fung, Survey of hallucination in natural language generation, *ACM Comput. Surv.* 55 (12) (2023) 1–38.
- [10] Y. Zhang, Y. Li, L. Cui, D. Cai, L. Liu, T. Fu, X. Huang, E. Zhao, Y. Zhang, Y. Chen, et al., Siren’s song in the AI ocean: a survey on hallucination in large language models, 2023, arXiv preprint [arXiv:2309.01219](https://arxiv.org/abs/2309.01219).
- [11] U. Khandelwal, O. Levy, D. Jurafsky, L. Zettlemoyer, M. Lewis, Generalization through memorization: Nearest neighbor language models, 2019, arXiv preprint [arXiv:1911.00172](https://arxiv.org/abs/1911.00172).
- [12] M. Li, R. Zhang, How far is language model from 100% few-shot named entity recognition in medical domain, 2023, arXiv preprint [arXiv:2307.00186](https://arxiv.org/abs/2307.00186).
- [13] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, et al., Retrieval-augmented generation for knowledge-intensive nlp tasks, *Adv. Neural Inf. Process. Syst.* 33 (2020) 9459–9474.
- [14] M. Li, L. Huang, Understand the dynamic world: An end-to-end knowledge informed framework for open domain entity state tracking, 2023, arXiv preprint [arXiv:2304.13854](https://arxiv.org/abs/2304.13854).
- [15] K. Taunk, S. De, S. Verma, A. Swetapadma, A brief review of nearest neighbor algorithm for learning and classification, in: 2019 International Conference on Intelligent Computing and Control Systems, ICCS, IEEE, 2019, pp. 1255–1260.
- [16] A. Dubey, A. Jauhri, A. Pandey, A. Kadian, A. Al-Dahle, A. Letman, A. Mathur, A. Schelten, A. Yang, A. Fan, et al., The llama 3 herd of models, 2024, arXiv preprint [arXiv:2407.21783](https://arxiv.org/abs/2407.21783).
- [17] M. Li, Z. Zhan, H. Yang, Y. Xiao, J. Huang, R. Zhang, Benchmarking retrieval-augmented large language models in biomedical NLP: Application, robustness, and self-awareness, 2024, arXiv preprint [arXiv:2405.08151](https://arxiv.org/abs/2405.08151).
- [18] M. Liu, Z. Tu, T. Su, X. Wang, X. Xu, Z. Wang, Behaviormet: A fine-grained behavior-aware network for dynamic link prediction, *ACM Trans. the Web* 18 (2) (2024) 1–26.
- [19] Y. Zhuang, Y. Yu, K. Wang, H. Sun, C. Zhang, Toolqa: A dataset for llm question answering with external tools, *Adv. Neural Inf. Process. Syst.* 36 (2024).
- [20] F. Jiang, C. Qin, K. Yao, C. Fang, F. Zhuang, H. Zhu, H. Xiong, Enhancing question answering for enterprise knowledge bases using large language models, 2024, arXiv preprint [arXiv:2404.08695](https://arxiv.org/abs/2404.08695).
- [21] K. Singhal, S. Azizi, T. Tu, S.S. Mahdavi, J. Wei, H.W. Chung, N. Scales, A. Tanwani, H. Cole-Lewis, S. Pfohl, et al., Large language models encode clinical knowledge, *Nature* 620 (7972) (2023) 172–180.
- [22] Y. Labrak, A. Bazoge, E. Morin, P.-A. Gourraud, M. Rouvier, R. Dufour, BioMistral: A collection of open-source pretrained large language models for medical domains, 2024, arXiv preprint [arXiv:2402.10373](https://arxiv.org/abs/2402.10373).
- [23] T. Han, L.C. Adams, J.-M. Papaioannou, P. Grundmann, T. Oberhauser, A. Löser, D. Truhn, K.K. Bressem, MedAlpaca—an open-source collection of medical conversational ai models and training data, 2023, arXiv preprint [arXiv:2304.08247](https://arxiv.org/abs/2304.08247).
- [24] M. Li, A. Blaes, S. Johnson, H. Liu, H. Xu, R. Zhang, Cancerllm: A large language model in cancer domain, 2024, arXiv preprint [arXiv:2406.10459](https://arxiv.org/abs/2406.10459).
- [25] K. Guu, K. Lee, Z. Tung, P. Pasupat, M. Chang, Retrieval augmented language model pre-training, in: International Conference on Machine Learning, PMLR, 2020, pp. 3929–3938.
- [26] L. Gao, Z. Dai, P. Pasupat, A. Chen, A.T. Chaganty, Y. Fan, V. Zhao, N. Lao, H. Lee, D.-C. Juan, et al., Rarr: Researching and revising what language models say, using language models, in: Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), 2023, pp. 16477–16508.

- [27] X. Li, R. Zhao, Y.K. Chia, B. Ding, L. Bing, S. Joty, S. Poria, Chain of knowledge: A framework for grounding large language models with structured knowledge bases, 2023, arXiv preprint [arXiv:2305.13269](https://arxiv.org/abs/2305.13269).
- [28] O. Ram, Y. Levine, I. Dalmedigos, D. Muhlgay, A. Shashua, K. Leyton-Brown, Y. Shoham, In-context retrieval-augmented language models, 2023, arXiv preprint [arXiv:2302.00083](https://arxiv.org/abs/2302.00083).
- [29] W. Shi, S. Min, M. Yasunaga, M. Seo, R. James, M. Lewis, L. Zettlemoyer, W.-t. Yih, Replug: Retrieval-augmented black-box language models, 2023, arXiv preprint [arXiv:2301.12652](https://arxiv.org/abs/2301.12652).
- [30] M. Li, H. Zhou, H. Yang, R. Zhang, RT: a retrieving and chain-of-thought framework for few-shot medical named entity recognition, *J. Am. Med. Informat. Assoc.* (2024) ocae095.
- [31] M. Li, M. Chen, H. Zhou, R. Zhang, Petailor: Improving large language model by tailored chunk scorer in biomedical triple extraction, 2023, arXiv preprint [arXiv:2310.18463](https://arxiv.org/abs/2310.18463).
- [32] O. Taboureau, S.K. Nielsen, K. Audouze, N. Weinhold, D. Edsgård, F.S. Roque, I. Kouskoumvekaki, A. Bora, R. Curpan, T.S. Jensen, et al., ChemProt: a disease chemical biology database, *Nucleic Acids Res.* 39 (suppl_1) (2010) D367–D372.
- [33] C. Sun, Z. Yang, L. Wang, Y. Zhang, H. Lin, J. Wang, Mrc4bioer: joint extraction of biomedical entities and relations in the machine reading comprehension framework, *J. Biomed. Inform.* 125 (2022) 103956.
- [34] I. Segura-Bedmar, P. Martínez Fernández, M. Herrero Zazo, Semeval-2013 task 9: Extraction of drug-drug interactions from biomedical texts (ddiextraction 2013), Association for Computational Linguistics, 2013.
- [35] H. Gurulingappa, A.M. Rajput, A. Roberts, J. Fluck, M. Hofmann-Apitius, L. Toldo, Development of a benchmark corpus to support the automatic extraction of drug-related adverse effects from medical case reports, *J. Biomed. Inform.* (ISSN: 1532-0464) 45 (5) (2012) 885–892, <http://dx.doi.org/10.1016/j.jbi.2012.04.008>, Text Mining and Natural Language Processing in Pharmacogenomics, URL <http://www.sciencedirect.com/science/article/pii/S1532046412000615>.
- [36] S. Kok, P. Domingos, Statistical predicate invention, in: Proceedings of the 24th International Conference on Machine Learning, 2007, pp. 433–440.
- [37] Y. Xiao, Y. Hou, H. Zhou, G. Diallo, M. Fiszman, J. Wolfson, L. Zhou, H. Kilicoglu, Y. Chen, C. Su, et al., Repurposing non-pharmacological interventions for alzheimer's disease through link prediction on biomedical literature, *Scientific reports* 14 (1) (2024) 8693.
- [38] B. King, J. Flanigan, Diverse retrieval-augmented in-context learning for dialogue state tracking, 2023, arXiv preprint [arXiv:2307.01453](https://arxiv.org/abs/2307.01453).
- [39] H. Touvron, T. Lavril, G. Izacard, X. Martinet, M.-A. Lachaux, T. Lacroix, B. Rozière, N. Goyal, E. Hambro, F. Azhar, et al., Llama: Open and efficient foundation language models, 2023, arXiv preprint [arXiv:2302.13971](https://arxiv.org/abs/2302.13971).
- [40] J. Devlin, M.-W. Chang, K. Lee, K. Toutanova, Bert: Pre-training of deep bidirectional transformers for language understanding, 2018, arXiv preprint [arXiv:1810.04805](https://arxiv.org/abs/1810.04805).
- [41] J. Lee, W. Yoon, S. Kim, D. Kim, S. Kim, C.H. So, J. Kang, Biobert: a pre-trained biomedical language representation model for biomedical text mining, *Bioinformatics* 36 (4) (2020) 1234–1240.
- [42] X. Yang, A. Chen, N. PourNejatian, H.C. Shin, K.E. Smith, C. Parisien, C. Compas, C. Martin, M.G. Flores, Y. Zhang, et al., GatorTron: a large clinical language model to unlock patient information from unstructured electronic health records, 2022, arXiv preprint [arXiv:2203.03540](https://arxiv.org/abs/2203.03540).
- [43] W. Tang, B. Xu, Y. Zhao, Z. Mao, Y. Liu, Y. Liao, H. Xie, UniRel: Unified representation and interaction for joint relational triple extraction, 2022, arXiv preprint [arXiv:2211.09039](https://arxiv.org/abs/2211.09039).
- [44] Y.-M. Shang, H. Huang, X. Mao, Onerel: Joint entity and relation extraction with one module in one step, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 36, no. 10, 2022, pp. 11285–11293.
- [45] C. Gao, W. Zhang, W. Lam, L. Bing, Easy-to-hard learning for information extraction, 2023, arXiv preprint [arXiv:2305.09193](https://arxiv.org/abs/2305.09193).
- [46] H. Tan, Z. Yang, Z. Ding, Z. Wang, L. Luo, L. Wang, Y. Zhang, W. Liu, H. Lin, J. Wang, Joint biomedical entity and relation extraction with unified interaction maps, in: 2023 IEEE International Conference on Bioinformatics and Biomedicine, BIBM, IEEE, 2023, pp. 1437–1442.
- [47] C. Yang, J. Deng, X. Chen, Y. An, SPBERE: Boosting span-based pipeline biomedical entity and relation extraction via entity information, *J. Biomed. Inform.* 145 (2023) 104456.
- [48] X. Zeng, S. He, D. Zeng, K. Liu, S. Liu, J. Zhao, Learning the extraction order of multiple relational facts in a sentence with reinforcement learning, in: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP), 2019, pp. 367–377.
- [49] J. Achiam, S. Adler, S. Agarwal, L. Ahmad, I. Akkaya, F.L. Aleman, D. Almeida, J. Altenschmidt, S. Altman, S. Anadkat, et al., Gpt-4 technical report, 2023, arXiv preprint [arXiv:2303.08774](https://arxiv.org/abs/2303.08774).